DEPARTMENT OF CSE

YEARLY QUESTION

ON

ELE TROCHEMISTRY

1. what is equivalence conductance?Why equivalence conductance of a weak electrolyte increases more rapidly than strong electrolyte? 3 (2019,

2. Distinguish between galvanic cell and electrolytic cell? Write the cell reaction for the following electrochemical cell:

i)$Zn|Zn^{2+}||Cu^{2+}|Cu$ ii) $Cd|Cd^{2+}||Hg_2Cl_2|Hg$; iii) $Pb|PbSO4(s)|SO_4^{-2}||Cu^{2+}|Cu$ 3 (2019,13

3.What is transference number?. 0.75 (2019,12,14

4.What is ionic mobility? State and explain Kohlrausch law of independent migration of ion 3 (2019,12

5. Calculate the EMF of the cell:$Zn|Zn^{2+}(0.001)||Ag^{+}(0.1)|Ag$. The standard potential of $Ag^{+}|Ag$ is 0.08v and $Zn|Zn^{2+}$ is -0.76 v 2.75/1.75 (2019,15

6.Define the term :liquid junction potential,electrode potential ,standard electrode potential, and concentration cell. 3 (2019

7.Differentiate "Electrolytic cell and electrochemical cell" 2 (2018

8.What happens when a copper rod is dipped in a zinc sulphate solution? 1.75 (2018

9.The galvanic cell represented s

$Mg(s)|Mg^{2+}(1mol.dm^{-3})||H^{+}(1mol.dm^{-3})|H_2(g)|Pt(s)$

And the reading found on the voltameter is 2.36v

Calculate the standard reduction potential of the magnesium hal-cell.Show all of your working .And the balanced NET(overall) cell reaction that take places in this cell.No spector ions are required . 2 (2018

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Debine: The term

Electrode, Anode, cathode, Electrolytic cell,

electrochemical cell, Electrolyte, strong electrolyte

weak electrolyte, conductance, specific conductance

Equivalence conductance.

Electrolysis

law of Faraday

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10.Why the specific conductance of a weak electrolyte decrease rapidly with increase in concentration 2 (2017

11.Define the term:i)Electromotive force ii) Equivalent conductance iii) cell constant 1.5 (2017

12.What current strength in ampere will be required to liberate 30 g iodine from potassium iodide solution in one hour? 2.25 (2017

13. Describe the calomel reference electrode and give half cell reaction when it is coupled with a standard hydrogen electrode ? 3. (2017,15

14. State Farady's law of electrolysis.Describe experiments by which the first law can be verified . 3+2 (2016

15 State and.At 25^0 c the values of equivalence conductance at infinite dilution of NaCl ,HCl ,CH_3COONa are 126.45 ,426.16, 91.0 $Cm^2 equi^{-1}$ respectively calculate the equivalence conductance of acetic acid at infinite dilution . 1.75 (2016

16.explain the principle of independent ion migration. 2 (2016

17.What is meant by molar conductance ?Describe a method of determining molar conductance of electrolyte . 1+3 (2015

18.What is cell constant ? How is it determined? 4.75 (2014

19. State and explain the principle of independent ion migration and mobility. Explain the relation $\lambda_a^0 + \lambda_c^0 = A^0$ 4 (2014

20. Describe one method for the determination of transport number

21.Write Debye-Huckel limiting law . 1 (2012

22.Will the transport number of Cl^- ino be the same in HCl,LiCl,KCl?If not explain how the transport number will change and why? 4 (2014

23. What is molar conductance?How does molar conductance vary with electrolyte concentration for a) strong electrolytes b) for weak electrolytes ? 3.75 (2013

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering **Part-I Odd** Semester Examination **2020**
Course Code: **CHEM1121** Course Title: **Chemistry**
(Session: **2019-2020**)
Time: **03** Hours Full Marks: **52.5**

[N.B. Answer 06 (Six) questions taking any 03 (Three) questions from each section. The figures in the right margin indicate full marks.]

Section A

1. (a) What is an electrolyte? Why does table salt solution conduct electricity, while sugar solution doesn't? 1+2
   (b) State the Faraday's Law of electrolysis. Mention its importance. 3
   (c) 0.1978g of copper is deposited by a current of 0.2 ampere in 50 minutes. What is the electrochemical equivalent of copper? 2.75

2. (a) Define specific conductance and give its unit. 2.75
   (b) 'On progressive dilution, specific conductance of an electrolyte decreases but molar conductance increases' discuss. 3
   (c) Explain briefly the Debye-Huckel theory for mean activity coefficient. 3

3. (a) What is a reference electrode? Describe the construction and working of standard hydrogen electrode. 1+2
   (b) Zn gives H_2 gas when it reacts with H_2SO_4 but Ag does not. Explain (given that $E^0{}_{Zn^{2+}/Zn} = -0.76\text{V}$ and $E^0{}_{Ag^+/Ag} = 0.80\text{V}$). 2.75
   (c) Discuss charge and discharge chemistry of Lead-acid storage battery. 3

4. (a) Write Rutherford's experiment of scattering of α-particles and give the drawbacks of the model. 3.75
   (b) State the postulates of Bohr theory of Hydrogen atom. 3
   (c) The energy transition in hydrogen atom occurs from n = 3 to n = 2 energy level. Calculate the wavelength of the emitted electron. 2
   Given that, $R = 1.097 \times 10^7\ m^{-1}$.

Section B

5. (a) What are ionic, covalent, and molecular crystals? Give examples. 3
   (b) Explain the term "delocalize electrons" in metal. 1.75
   (c) Draw a unit cell of diamond crystal. Why diamond is non-conductor? 2
   (d) Draw the layer structure of Graphite. Mention the hybridization and force belongs in that structure. 2

6. (a) What is atomic mass of an Isotope? Why are atomic masses of most of the isotopes of a given element fractional and not whole number? 3.75
   (b) What do you mean by natural radioactivity and artificial radioactivity? 2
   (c) If you are given a piece of mineral, how can you determine by simple experiments if the mineral is radioactive? 2
   (d) Write a reaction for α-decay. 1

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7. (a)	What do you understand by hydrogen bonds? Classify them with examples.	2
(b)	In what way metals and ionic salts conduct electricity? Explain with suitable examples.	3
(c)	When does a bonding in a covalent compound become fully ionic?	1.75
(d)	Discuss the type of hybridization that exist in tetrahedral molecule.	2
8. (a)	Define transition elements. Why do transition elements show variable oxidation states?	1+1.5
(b)	What are the everyday uses of transition metals?	1.5
(c)	Compare the chemistry of the lanthanides with that of the actinides.	2.5
(d)	What is meant by "disproportionation" of oxidation state? Why Cu+ is not stable in aqueous solution?	2.25

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DEPARTMENT OF CSE

YEARLY QUESTION

ON

Periodic Table

1.What are s and p block elements?Mention their five characterstics. 3 (2018,16

2.Define electron affinity? Electron affinity of F is less than that of Cl./Cl has largest value of electron affinity . (2019,17

Why 2.75 (2019,17

3.Explain why 6 (2018,2017

i)Flourine is more electronegative than chlorine.

ii)Na+ is smaller than Na but Cl- is larger than Cl-atom/ Cation are smaller than their parents atom

iii)The first ionization energy of N is more than that of O ./ the ionization potential of Al is lower than that of Mg (2018,2016,14

iv)The electron affinity of noble gases is zero.

4. Draw the periodic table for first 36 elements with their correct symbols.you should also label of groups and periods of that segments table . 5 (2018

5.Write down the electronic configuration of Chrominium and Technetium. 2 (2018

6. Find effective nucler charge acting on one of the 4s electrons of V(23) 2017

7. What is periodic law?Write down the classification of elements on the basis of electronic configuration 3.75 (2017,13

8.How does atomic radius change as you move a)from left to right across the period b)from top to bottom in a group 2 (2015,14

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9.What is meant by electron affinity? Arrange the following elements in order of their increasing electron affinity: F,Na,Cl,K,Br,and I 2 (2015,2012

10.What is covalent radius? Na^+ is smaller in size than F^-, though both are isoelectrionic 2.5 (2013

11.Define electron affinity? Explain why the electron affinity of F- atom is higher than that of O-atom 2.5 (2013

12.Arrange the following elements in their increasing order of their first ionization energy .

i)Li,Be,B ii) N,O,F iii)C,N,O,F 3 (2012

Li L Be L B ↓

i) Li L Be L B

ii) O N

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DEPARTMENT OF CSE

YEARLY QUESTION

ON

ATOMIC STRUCTURE

1.Write down the postulates of Bohr's atomic model. 2 (2015,2013 ,2020

2.Discuss the drawbacks/inadequacies of Rutherford's model for atomic model structure. 1.5/2 (2018,2016,2012

3.State the limitations of Bohr's atomic model. 1.5 (2018 , 2020

4.Explain with examples Hund's rule of maximum multiplicity.How does this help in determining the electronic configuration of elements. 2.75 (2018,

5.Derive an expression for the radius for the radius of n-th orbit of H-atom 3.75

(2019,2017,2016,2014,2012

6.Calculate the radii of fourth orbit in H-atom. Given that $h=6.626\times10^{-27}$ erg-s,$m=9.1\times10^{-28}$ g $\pi=3.14$,$e=4.8\times10^{-10}$ esu

7.Calculate the radius of second orbit He^+ ion 1.5 (2019

8.Calculate the radius of 1st orbit He^+ ion .Given that Bohr radius is 0.529×10^{-10} m 2 (2014,

9.Find out the four quantum numbers for the 15th electron of Cl-atom 1.75(2017

10.State and explain the following term:Aufbau principle,Hund's principle,Pauli's exclusion principle, Heisenberg uncertainty principle . 02
(2019,2018,2017,2016,2013

11.How many subshells are possible in third energy shell?what is the maximum number of electrons that can occupy a d subshell? 2 (2013

12.What are quantum numbers?what is the subshell designation for each of the following cases?

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i)n=2,l=0;ii)n=5,l=1;iii)n=4,l=3 1.75 (2016,2014

13.What is (n+l) rule or Why the last electron k(19) goes to 4s orbital instead of 3d orbital 01(2015Which of the following orbital are possible or not possible 1P,2S,2d,3f 1.75 (2017,2013

14.Write down De-Broglie equation or χ=

15.Calculate the wave length in A^0 of the line in Balmer series that is associated with the drop of the electron from the fouth orbit? $R_{H} = 109{,}676\ cm^{-1}$ 2.75

16. Calculate the frequency in term of wavelength for second spectral line of lyman series.Given that $R_H = 1.097\times10^7 m^{-1}$ 1.5(2019,2016

17.Calculate the frequency of electromagnetic radiation emitted by hydrogen atom in the electron transition from n=4 to n=3 . Given that $R_H = 1.097\times10^7 m^{-1}$ and $h=6.626\times10^{-34}$ Js 2 (2015 , 2020

18.The n quantum number of an atomic orbital is 6.What are the possible values of l? what are the possible value of m_l if l quantum number is 5? 1.75 (2015

19.State whether each of the following sets of quantum numbers is permissible for an electron in an atom.If a set is not permissible, explain why. 1 (2015

(i) (1,0,+1/2,+1/2) and (ii) (3,2,1,1)

20. write down Rutherford's experiment of scattering α-particles and give the drawbacks of the model.

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